

# Answer Key — Biology Grade 11

Sub-Strand 2.1: Reproduction in Plants • Lesson 3: Why Can a Lone Pawpaw Tree Flower Heavily but Never Set Fruit? Adaptations That Prevent Self-Pollination

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 | Q8 | Q9 |
|----|----|----|----|----|----|----|----|----|
| A  | B  | B  | D  | B  | D  | C  | C  | D  |

USE AFTER: PREDICT PHASE — Predict: photo of a lone pawpaw/avocado tree covered in flowers with no fruit

**Q1. A pawpaw tree stands alone in a compound in Kisumu. It has hundreds of flowers every season but never sets a single fruit. Which explanation is best supported by what we know about pawpaw?**

- A. Pawpaw trees are usually either male or female, so a lone tree may have no partner tree to supply pollen or ovules**
- B. Pawpaw flowers always pollinate themselves, so the tree must be sick
- C. Pawpaw trees only produce fruit when grown from cuttings, not from seed
- D. The tree has too many flowers, so none can develop into fruit

**Correct answer: A.** Pawpaw shows dioecy: male and female flowers are on different trees. A lone tree of one sex cannot pollinate itself, so flowers form but no fruit follows.

USE AFTER: OBSERVE PHASE — Station 1 - Timing (protandry/protogyny): maize tassels and silks timeline

**Q2. In a maize plant, the tassels shed their pollen before the silks on the same plant have emerged and are ready to receive pollen. What is this timing adaptation called?**

- A. Dioecy
- B. Protandry**
- C. Protogyny
- D. Heterostyly

**Correct answer: B.** Protandry is when the male parts release pollen before the female parts of the same plant are receptive. This reduces self-pollination.

USE AFTER: OBSERVE PHASE — Station 2 - Structure (heterostyly): wire/bead pin and thrum flower models

**Q3. A 'pin' flower has a long style and short stamens. A 'thrum' flower has a short style and long stamens. How does this arrangement (heterostyly) promote cross-pollination?**

- A. The stigma of each flower chemically rejects its own pollen
- B. The anthers and stigma are at different heights, so an insect carries pollen from a flower of one type to the stigma of the other type**
- C. The pollen is shed before the stigma is ready
- D. The flowers are on separate trees, so pollen must travel between trees

**Correct answer: B.** In pin and thrum flowers the anthers and stigma sit at different heights. Pollen from a flower's own anther is poorly placed to reach its own stigma, while an insect moves pollen between the two types.

USE AFTER: OBSERVE PHASE — Station 3 - Separate sexes (dioecy): comparison of pawpaw with maize (monoecious)

**Q4. Maize has separate male flowers (tassels) and female flowers (cobs with silks) on the same plant. Does this arrangement completely prevent self-pollination?**

- A. Yes, because each maize plant is either male or female
- B. Yes, because male and female flowers can never meet
- C. No, because maize flowers have both male and female parts in the same flower
- D. No, pollen from the tassel can still land on silks of the same plant, so self-pollination is only reduced**

**Correct answer: D.** Maize is monoecious: separate flowers on one plant. Pollen can still fall or blow onto silks of the same plant, so self-pollination is reduced but not eliminated. Dioecy, with separate male and female plants as in pawpaw, is different.

USE AFTER: OBSERVE PHASE — Station 4 - Chemistry (self-incompatibility): passion fruit coloured-bead simulation

**Q5. A passion fruit flower has mature pollen and a receptive stigma. Pollen from the same flower lands on its stigma, but the pollen does not germinate and is rejected. Which adaptation is this?**

- A. Protandry
- B. Self-incompatibility**
- C. Dioecy
- D. Heterostyly

**Correct answer: B.** In self-incompatibility the stigma recognises and rejects pollen from the same plant, even when both are mature. Timing, height and separate sexes are not involved here.

USE AFTER: EXPLAIN PHASE — Part A - Think-Pair-Share: which adaptation explains the lone pawpaw and why cassava cuttings need no partner

**Q6. A farmer in Kisii plants cassava cuttings and gets new plants without any flowers or partner plants. Why do cassava cuttings not need a partner?**

- A. Cassava cuttings produce pollen that fertilises its own ovules
- B. Cassava trees are male or female, so cuttings take the correct sex
- C. Cassava has self-incompatibility, so it avoids partners
- D. Growing from cuttings is asexual reproduction, so there is no pollination and the new plant is identical to the parent**

**Correct answer: D.** Cuttings grow from a piece of stem, so no pollen, ovule or fertilisation is needed. The new plant is a clone of the parent, so the barriers to self-pollination do not matter.

USE AFTER: EXPLAIN PHASE — Part B - Claim-Evidence-Reasoning: reasoning that adaptations promote cross-pollination and genetic variation

**Q7. Which statement best explains why many flowering plants have adaptations that promote cross-pollination?**

- A. Flowers that self-pollinate are weak and always die
- B. Self-pollination never produces seeds in any plant
- C. Cross-pollination mixes the genes of two parents, giving varied offspring that may resist disease better**
- D. Cross-pollination produces offspring that are exact copies of the parent

**Correct answer: C.** Cross-pollination brings together genes from different plants, which gives genetic variation. This is useful to the species, for example varied maize crops resisting disease. It is a problem only for a farmer with a single tree.

USE AFTER: DQB PHASE — Driving Question Board: 'What we figured out' and 'New questions' sticky notes

**Q8. After today's work, a class says they can now explain why a lone tree full of flowers bears no fruit. Which new question follows most logically from their evidence?**

- A. Why does a lone tree have flowers but no fruit?
- B. Why can a cassava cutting grow into a whole plant without a partner?
- C. What happens inside the flower after compatible pollen lands on a receptive stigma?**
- D. Why are maize tassels and silks on the same plant?

**Correct answer: C.** The other three questions were already answered by the station evidence. The step after compatible pollen lands on the stigma has not yet been explained, so it is the new question for the next lesson.

USE AFTER: END OF LESSON — Model revision: farmers' choices and the lone pawpaw case (partner tree needed)

**Q9. A farmer has one female pawpaw tree. It flowers well but gives no fruit. What should the farmer do to get fruit?**

- A. Spray the tree with water so the stigmas can reject their own pollen
- B. Take cuttings from the female tree and plant them next to it
- C. Cut off half of the flowers so the rest can pollinate themselves
- D. Plant a male pawpaw tree nearby so pollen can be carried to the female flowers**

**Correct answer: D.** Pawpaw is dioecy, so female flowers need pollen from a male tree. Cuttings from the female tree would also be female. Pollinating itself is impossible for a female tree.